

Total No. of Questions : 10]

SEAT No. :

P3106

[5154]-672

[Total No. of Pages : 3

B.E.(Computer Engineering)
PRINCIPLES OF MODERN COMPILER DESIGN
(2012 Pattern) (Semester-I) (410442)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) figures to the right indicate full marks.

Q1) a) Write down the regular expression for the following **[4]**

- i) Comment in C.
- ii) Floating point number.

b) Write a Syntax directed translation scheme for Boolean Expression. **[6]**

OR

Q2) a) Consider the statement: **[4]**

$X[i, j] := Y[i+j, k] + z.$

The maximum dimensions of X are [d1,d2] and of Y are [d3,d4].

Generate three address code.

b) What are synthesized and inherited attributes? What are Marker Non terminal symbols? Give example. **[6]**

Q3) a) Write a short note on I/P buffering used in Lexical Analyzer. **[4]**

b) Check whether the following grammar LL(1) or not. **[6]**

$E \rightarrow TE'$

$E' \rightarrow *TE' / \epsilon$

$T \rightarrow FT'$

$T' \rightarrow ^T / \epsilon$

$F \rightarrow (E) / id$

P.T.O.

OR

Q4) a) What is need of Semantic Analysis? Explain the position of Type Checker with diagram. [4]

b) Show that the following grammar is not SLR (1) [6]

$S \rightarrow Aa Ab|B b Ba$

$A \rightarrow \epsilon$

$B \rightarrow \epsilon$

Q5) a) Write a note on application of Directed Acyclic Graph (DAG) in code generation. [6]

b) Write an algorithm for copy propogation. [6]

c) Write a short note on Data flow equations and iterative data flow analysis. [6]

OR

Q6) a) Describe in detail about a simple code generator with the appropriate algorithm. [6]

b) Discuss about the following: [6]

i) Dead-code Elimination and

ii) Code motion.

c) Show the steps involved on generating the code for the expression: [6]

$(x+y)/(p+q)$

Q7) a) Discuss source language issues related to Object Oriented languages. [6]

b) Explain code generation for control flow statements. [6]

c) Explain Polymorphic typing with respect to Functional languages. [4]

OR

Q8) a) Explain following related to Haskell program. [6]

i) Offside rule.

ii) Lists.

- b) Explain following with respect to Functional languages. [6]
- i) Referential transparency.
 - ii) Lazy evaluation.
- c) What is activation record? Explain possible structure of an activation record? [4]
- Q9)** a) Discuss the issues in Tuple Space implementation. [6]
- b) Write short notes on [6]
- i) JIT
 - ii) nmake
- c) Explain following shared variable models [4]
- i) Locks
 - ii) Monitors

OR

- Q10)** a) Explain cross compilation using XMLVM. [6]
- b) Discuss following with respect to Parallel object oriented languages. [6]
- i) Object location
 - ii) Object migration
- c) What is interpreter? Explain JVM interpreter. [4]

❄ ❄ ❄